Strand 3: Geometry and Measurement

Sub-strand 2: Measurement and Vectors

Lesson 2: Position Vectors, Vector Arithmetic, and Component Equations

1. Curriculum Standard and Overview

- **Strand**: 3 — Geometry and Measurement
- **Sub-strand**: 2 — Measurement and Vectors
- **Content Standard**: B9.3.2.1 / B9.3.2.2
- **Indicators**:
- **B9.3.2.1.4** — Express points in the Cartesian plane as position vectors.
- **B9.3.2.2.2** — Apply the triangular and parallelogram laws of addition to resolve vectors.
- **Mock-paper links**: P1 Q13 (find B given A and `\vec{AB}`), P1 Q21 (find `\vec{BC}`), P1 Q38 (equate vector components), P2 Q6(d) (vector addition with unknowns).

2. Preamble — What Is a Vector?

A **vector** is a quantity with **size and direction**.

- Displacement "3 km North" is a vector.
- Speed "10 km/h" is a **scalar** (size only). Velocity "10 km/h East" is a vector.

We write a 2-D vector as a **column**:

$$\vec{v} = \begin{pmatrix} x \\ y \end{pmatrix}$$

- x = horizontal component (east–west)
- y = vertical component (north–south)

3. Position Vectors

The **position vector** of a point $P(x, y)$ is the vector from the origin $O(0,0)$ to P:

$$\vec{OP} = \begin{pmatrix} x \\ y \end{pmatrix}$$

Displacement between two points:

$$\vec{AB} = \vec{OB} - \vec{OA} = \begin{pmatrix} x_B - x_A \\ y_B - y_A \end{pmatrix}$$

Mock P1 Q13

A is $(2, 3)$ and $\vec{AB} = \begin{pmatrix} 5 \\ -4 \end{pmatrix}$. Find B.

$$\vec{OB} = \vec{OA} + \vec{AB} = \begin{pmatrix} 2 \\ 3 \end{pmatrix} + \begin{pmatrix} 5 \\ -4 \end{pmatrix} = \begin{pmatrix} 7 \\ -1 \end{pmatrix}$$

Answer: A — $(7, -1)$**.**

4. Vector Addition and Subtraction

4.1 Adding components

$$\begin{pmatrix} a \\ b \end{pmatrix} + \begin{pmatrix} c \\ d \end{pmatrix} = \begin{pmatrix} a+c \\ b+d \end{pmatrix}$$

4.2 Triangle Law

$$\vec{AB} + \vec{BC} = \vec{AC} \Rightarrow \vec{BC} = \vec{AC} - \vec{AB}$$

Mock P1 Q21

$\vec{AB} = \begin{pmatrix} 3 \\ 5 \end{pmatrix}$, $\vec{AC} = \begin{pmatrix} -4 \\ 5 \end{pmatrix}$. Find $\vec{BC}$.

$$\vec{BC} = \vec{AC} - \vec{AB} = \begin{pmatrix} -4-3 \\ 5-5 \end{pmatrix} = \begin{pmatrix} -7 \\ 0 \end{pmatrix}$$

Answer: D — $\begin{pmatrix} -7 \\ 0 \end{pmatrix}$**.**

Mock P2 Q6(d)

$a = \begin{pmatrix} -4 \\ p \end{pmatrix}$, $b = \begin{pmatrix} q \\ 1 \end{pmatrix}$, $a + b = \begin{pmatrix} -1 \\ -8 \end{pmatrix}$. Find p and q.

Top: $-4 + q = -1 \Rightarrow q = 3$. **Bottom:** $p + 1 = -8 \Rightarrow p = -9$.

Answer: $p = -9$**,** $q = 3$**.**

Mock P1 Q38

$P = \begin{pmatrix} 3x-6 \\ 2y+5 \end{pmatrix}$, $R = \begin{pmatrix} 9 \\ 7 \end{pmatrix}$, $P = R$. Find x and y.

Top: $3x - 6 = 9 \Rightarrow 3x = 15 \Rightarrow x = 5$. **Bottom:** $2y + 5 = 7 \Rightarrow 2y = 2 \Rightarrow y = 1$.

Answer: C — $x = 5$ **and** $y = 1$**.**

5. Parallel, Perpendicular, and Magnitude (Preview)

- **Equal vectors:** same components.
- **Parallel:** $a = k\ b$ (components proportional).
- **Magnitude:** $|v| = \sqrt{(x^2 + y^2)}$.

6. Twenty Fully Worked Examples

Example 1 — Mock P1 Q13

$B = (7, -1)$. **A.**

Example 2 — Mock P1 Q21

$\vec{BC} = \begin{pmatrix} -7 \\ 0 \end{pmatrix}$. **D.**

Example 3 — Mock P2 Q6(d)

$q = 3$, $p = -9$.

Example 4 — Mock P1 Q38

$x = 5, y = 1$. **C.**

Example 5 — Position vector

$P(4, -3) \Rightarrow \vec{OP} = \begin{pmatrix} 4 \\ -3 \end{pmatrix}$.

Example 6 — Add vectors

$\begin{pmatrix} 2 \\ 3 \end{pmatrix} + \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 3 \\ 2 \end{pmatrix}$.

Example 7 — Subtract vectors

$\begin{pmatrix} 5 \\ 4 \end{pmatrix} - \begin{pmatrix} 2 \\ 6 \end{pmatrix} = \begin{pmatrix} 3 \\ -2 \end{pmatrix}$.

Example 8 — Displacement

$A(1, 2), B(4, 6)$: $\vec{AB} = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$.

Example 9 — Find A given B and $\vec{AB}$

$B(5, 5), \vec{AB} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$: $\vec{OA} = \vec{OB} - \vec{AB} = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$.

Example 10 — Triangle law

$\vec{AB} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \vec{BC} = \begin{pmatrix} 3 \\ -2 \end{pmatrix} \Rightarrow \vec{AC} = \begin{pmatrix} 4 \\ 0 \end{pmatrix}$.

Example 11 — Equate components

$\begin{pmatrix} 2a \\ b+1 \end{pmatrix} = \begin{pmatrix} 6 \\ 5 \end{pmatrix} \Rightarrow a = 3, b = 4$.

Example 12 — Scalar multiplication

$3\begin{pmatrix} 2 \\ -1 \end{pmatrix} = \begin{pmatrix} 6 \\ -3 \end{pmatrix}$.

Example 13 — Magnitude

$\begin{pmatrix} 3 \\ 4 \end{pmatrix}$: $|v| = \sqrt{(9+16)} = 5$.

Example 14 — Zero vector

$\begin{pmatrix} 0 \\ 0 \end{pmatrix}$.

Example 15 — Midpoint as vector

$A(0,0), B(6,8)$: midpoint position = $\begin{pmatrix} 3 \\ 4 \end{pmatrix}$.

Example 16 — Parallel vectors

$\begin{pmatrix} 2 \\ 4 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$ are parallel (one is 2× the other).

Example 17 — Word: displacement

Walk 3 km East, 4 km North: resultant = $\begin{pmatrix} 3 \\ 4 \end{pmatrix}$, magnitude 5 km.

Example 18 — Reverse displacement

$\vec{AB} = \begin{pmatrix} 2 \\ -3 \end{pmatrix} \Rightarrow \vec{BA} = \begin{pmatrix} -2 \\ 3 \end{pmatrix}$.

Example 19 — Unknown in displacement

$A(2, 1)$, $B(x, 5)$, $\vec{AB} = \begin{pmatrix} 3 \\ 4 \end{pmatrix} \Rightarrow x = 5$.

Example 20 — Combined

$a = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$, $b = \begin{pmatrix} -2 \\ 0 \end{pmatrix}$: $2a + b = \begin{pmatrix} 0 \\ 4 \end{pmatrix}$.

7. Comprehensive Exercise Sets (40 Questions)

All 4 authentic mock vector items are included. Generated questions top up to 40. Spiral review ≈ 15%.

Exercise 1 — Position vectors (includes mock)

1. **[Mock 2026 P1 Q13]** A is $(2, 3)$. If $\vec{AB} = \begin{pmatrix} 5 \\ -4 \end{pmatrix}$, find the coordinates of B.

A. $(7, -1)$ B. $(-3, 7)$ C. $(3, -7)$ D. $(5, 4)$

2. Write the position vector of $P(4, -3)$.
3. $A(1, 2)$, $B(4, 6)$. Find $\vec{AB}$.
4. $B(5, 5)$, $\vec{AB} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$. Find A.
5. **Spiral review:** If the area of a circle is 64π, find the circumference.

Exercise 2 — Vector addition & subtraction (includes mock)

1. **[Mock 2026 P1 Q21]** If $\vec{AB} = \begin{pmatrix} 3 \\ 5 \end{pmatrix}$ and $\vec{AC} = \begin{pmatrix} -4 \\ 5 \end{pmatrix}$, find $\vec{BC}$.

A. $\begin{pmatrix} 7 \\ 10 \end{pmatrix}$ B. $\begin{pmatrix} -7 \\ -10 \end{pmatrix}$ C. $\begin{pmatrix} 7 \\ -10 \end{pmatrix}$ D. $\begin{pmatrix} -7 \\ 0 \end{pmatrix}$

2. $\begin{pmatrix} 2 \\ 3 \end{pmatrix} + \begin{pmatrix} 1 \\ -1 \end{pmatrix} = ?$
3. $\begin{pmatrix} 5 \\ 4 \end{pmatrix} - \begin{pmatrix} 2 \\ 6 \end{pmatrix} = ?$
4. $\vec{AB} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$, $\vec{BC} = \begin{pmatrix} 3 \\ -2 \end{pmatrix}$. Find $\vec{AC}$.
5. **Spiral review:** A hall 20 m long is drawn as 10 cm. What is the scale?

Exercise 3 — Component equations (includes mock)

1. **[Mock 2026 P2 Q6(d)]** $a = \begin{pmatrix} -4 \\ p \end{pmatrix}$, $b = \begin{pmatrix} q \\ 1 \end{pmatrix}$, $a + b = \begin{pmatrix} -1 \\ -8 \end{pmatrix}$. Find p and q.
2. **[Mock 2026 P1 Q38]** $P = \begin{pmatrix} 3x-6 \\ 2y+5 \end{pmatrix}$, $R = \begin{pmatrix} 9 \\ 7 \end{pmatrix}$, $P = R$. Find x and y.

A. $x = 1, y = 5$ B. $x = -5, y = -1$ C. $x = 5, y = 1$ D. $x = -1, y = 5$

3. $\begin{pmatrix} 2a \\ b+1 \end{pmatrix} = \begin{pmatrix} 6 \\ 5 \end{pmatrix}$. Find a and b.
4. **Spiral review:** $13x - 12 = 5x + 60$. Find x.

5. **Spiral review:** Classify -3 under \mathbb{N}, \mathbb{W}, \mathbb{Z}, \mathbb{Q}.

Exercise 4 — Scalar multiplication & magnitude

1. 3\begin{pmatrix} 2 \\ -1 \end{pmatrix} = ?
2. Find the magnitude of \begin{pmatrix} 3 \\ 4 \end{pmatrix}.
3. \vec{AB} = \begin{pmatrix} 2 \\ -3 \end{pmatrix}. Find \vec{BA}.
4. **Spiral review:** Find 12x + 9q when x = 11, q = 3.
5. **Spiral review:** A van travels 154 km in 1(3)/(4) h. Speed?

Exercise 5 — Displacement problems

1. Walk 3 km East and 4 km North. Express the resultant as a vector and find its magnitude.
2. A(2, 1), B(x, 5), \vec{AB} = \begin{pmatrix} 3 \\ 4 \end{pmatrix}. Find x.
3. A(0,0), B(6,8). Find the position vector of the midpoint.
4. **Spiral review:** Area of a semicircle with r = 7 (π = 22/7).
5. **Spiral review:** (n)/(100) = 10.5. Find n.

Exercise 6 — Parallel & zero vectors

1. Are \begin{pmatrix} 2 \\ 4 \end{pmatrix} and \begin{pmatrix} 1 \\ 2 \end{pmatrix} parallel?
2. Write the zero vector.
3. a = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, b = \begin{pmatrix} -2 \\ 0 \end{pmatrix}. Find 2a + b.
4. **Spiral review:** 2x + 1 > 9. Solve.
5. **Spiral review:** Solve (-3)^(x-2) = -243.

Exercise 7 — Mixed vector arithmetic

1. \begin{pmatrix} 2 \\ 3 \end{pmatrix} + \begin{pmatrix} -1 \\ 4 \end{pmatrix} - \begin{pmatrix} 1 \\ 1 \end{pmatrix} = ?
2. \vec{AB} = \begin{pmatrix} 3 \\ -1 \end{pmatrix}, \vec{CD} = \begin{pmatrix} -3 \\ 1 \end{pmatrix}. What is the relationship?
3. Find k if k\begin{pmatrix} 2 \\ 6 \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \end{pmatrix}.
4. **Spiral review:** Three consecutive odd integers sum to 33.
5. **Spiral review:** Order (3)/(8), (1)/(2), (5)/(16).

Exercise 8 — Challenge

1. A = \begin{pmatrix} 2a \\ b-1 \end{pmatrix} and B = \begin{pmatrix} 8 \\ 3 \end{pmatrix} are equal. Find a and b.
2. A(1, 2), B(4, 6), C(0, 0). Find \vec{AB} + \vec{BC} + \vec{CA}.
3. If \vec{AB} = \begin{pmatrix} -7 \\ 0 \end{pmatrix} and A = (10, 3), find B.
4. **Spiral review:** Isosceles triangle XY = YZ, vertex 42^\circ. Find base angle.
5. **Spiral review:** Find the gradient through (1, 2) and (4, -4).

8. Full Step-by-Step Worked Solutions

Solutions to Exercise 1

1. B = (2+5, 3-4) = (7, -1) → **A**.
2. \begin{pmatrix} 4 \\ -3 \end{pmatrix}.
3. \begin{pmatrix} 3 \\ 4 \end{pmatrix}.

4. $A = (3, 4)$.
5. 16π.

Solutions to Exercise 2

1. $\begin{pmatrix} -4-3 \\ 5-5 \end{pmatrix} = \begin{pmatrix} -7 \\ 0 \end{pmatrix} \rightarrow$ **D**.
2. $\begin{pmatrix} 3 \\ 2 \end{pmatrix}$.
3. $\begin{pmatrix} 3 \\ -2 \end{pmatrix}$.
4. $\begin{pmatrix} 4 \\ 0 \end{pmatrix}$.
5. $1 : 200$.

Solutions to Exercise 3

1. $q = 3$, $p = -9$.
2. $x = 5$, $y = 1 \rightarrow$ **C**.
3. $a = 3$, $b = 4$.
4. $x = 9$.
5. ■, ■.

Solutions to Exercise 4

1. $\begin{pmatrix} 6 \\ -3 \end{pmatrix}$.
2. $|v| = 5$.
3. $\vec{BA} = \begin{pmatrix} -2 \\ 3 \end{pmatrix}$.
4. 159.
5. 88 km/h.

Solutions to Exercise 5

1. $\begin{pmatrix} 3 \\ 4 \end{pmatrix}$; magnitude 5 km.
2. $x = 5$.
3. $\begin{pmatrix} 3 \\ 4 \end{pmatrix}$.
4. 77 cm².
5. $n = 1050$.

Solutions to Exercise 6

1. **Yes** (one is 2× the other).
2. $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$.
3. $\begin{pmatrix} 0 \\ 4 \end{pmatrix}$.
4. $x > 4$.
5. $x = 7$.

Solutions to Exercise 7

1. $\begin{pmatrix} 0 \\ 6 \end{pmatrix}$.
2. **Opposite** (negatives of each other).
3. $k = (1)/(2)$.
4. **9, 11, 13**.
5. $(5)/(16) < (3)/(8) < (1)/(2)$.

Solutions to Exercise 8

1. $a = 4$, $b = 4$.
2. $\vec{AB} + \vec{BC} + \vec{CA} = \vec{0} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$.

3. `B = (10-7, 3+0) = (3, 3)`.
4. `69^\circ`.
5. `m = -2`.

9. Authentic Mock Items in This Lesson

Ref	Question	Answer
P1 Q13	`A(2,3)`, `\vec{AB} = (5,-4)`	A — `(7, -1)`
P1 Q21	`\vec{AB}=(3,5)`, `\vec{AC}=(-4,5)`	D — `(-7, 0)`
P1 Q38	`P = R` component equations	C — `x = 5`, `y = 1`
P2 Q6(d)	`a + b = (-1, -8)`	`p = -9`, `q = 3`

10. Lesson Summary

Skill	Rule
Position vector	`\vec{OP} = (x, y)`
Displacement	`\vec{AB} = \vec{OB} - \vec{OA}`
Triangle law	`\vec{AB} + \vec{BC} = \vec{AC}`
Component equation	equate tops and bottoms
Magnitude	`√(x^2 + y^2)`

Lesson prepared under `Grade9-Maths/Rules.md` *— plain English, British notation, full workings required.*